

in IPAbet keystrokes

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b p b		t d t d			ʈ ɖ ʈʀ ɖʀ	c ɟ c dʝ	k ɡ k ɡʁ	q ɢ q ɡʊ		ʔ ʔʰ
Nasal	m m	ɱ mʋ	n n			ɳ nʀ	ɲ nʝ	ŋ nɣ	ɴ nɔ		
Trill	ʙ rʙ		ʀ r						ʁ rɣ		
Tap or Flap		ⱱ 4v	ɾ 4ʰ			ɽ 4ʀ					
Fricative	ɸ β pʰ bʰ	f v f v	θ ð tʰ dʰ	s z s z	ʃ ʒ sʰ zʰ	ɬ ɮ sʀ zʀ	ç ʝ çH ɡʝ	x ɣ x ɣʰ	χ ʁ qʰ rʊ	ħ ʕ ʔʰ ʕʰ	h ɦ h hʰ
Lateral fricative			ɬ ɮ sʟ zʟ								
Approximant		ʋ vʰ	ɹ rʰ			ɻ rʀ	j j	ɰ wʰ			
Lateral approximant			l l			ɭ lʀ	ʎ lʝ	ʟ lɣ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

CONSONANTS (NON-PULMONIC)

Clicks

- **p** Bilabial
- | **t** Dental
- ! **q** (Post)alveolar
- ‡ **c** Palatoalveolar
- || **l** Alveolar lateral

Voiced implosives

b **bP** Bilabial
 d **dP** Dental/alveolar
 ʃ **jP** Palatal
 ɡ **gP** Velar
 ɣ **qP** Uvular

Ejectives

p'  Bilabial
t'  Dental/alveolar
k'  Velar
s'  Alveolar fricative

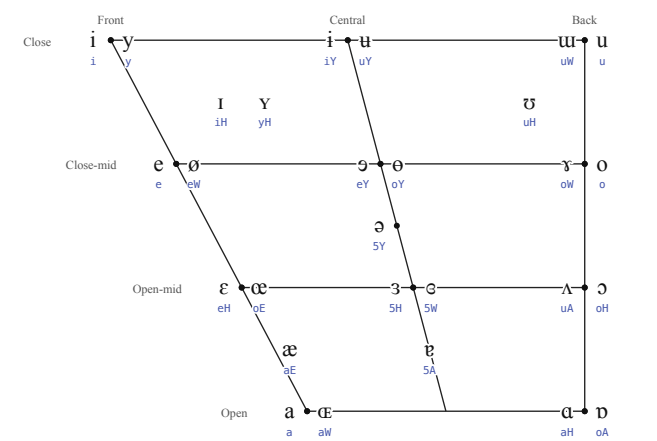
OTHER SYMBOLS

M		Voiceless labial-velar fricative	ɕ		Voiceless alveolo-palatal fricative
W		Voiced labial-velar approximant	ʐ		Voiced alveolo-palatal fricative
ɰ		Voiced labial-palatal approximant	ɭ		Voiced alveolar lateral flap
H		Voiceless epiglottal fricative	ʈʂ		Affricate (tie bar)
ʕ		Voiced epiglottal fricative	ɸ		Simultaneous f and x
ʡ		Epiglottal plosive			

DIACRITICS

η [vɤk] Voiceless	ḡ [vɤu] Breathy voiced	ʈ [vt] Dental
$\S$ [vɤv] Voiced	ḥ [vɤn] Creaky voiced	ʈ̪ [vd] Apical
$^{\text{h}}$ [vɤz h] Aspirated	ʈ̺ [vl] Linguolabial	ʈ̟ [vɤd] Laminal
ɔ̹ [vɤw] More rounded	w [vɤz w] Labialized	$\tilde{\epsilon}$ [vn] Nasalized
ɔ̜ [vɤw̥] Less rounded	j [vɤz j] Palatalized	n [◌] [vɤz n] Nasal release
ʉ [vɤ=] Advanced	ɣ [vɤz g ɤH] Velarized	l [◌] [vɤz l] Lateral release
ɛ̠ [vɤ=] Retracted	ʕ [vɤz 3 ɤH] Pharyngealized	ḍ [vp] No audible release
ẽ [vɤu] Centralized	ɫ [vɤy] Velarized or pharyngealized	
ẽ̞ [vɤx] Mid-centralized	ɇ [vɤg] Raised	
η [vɤs] Syllabic	ɇ̞ [vg] Lowered	
ɛ̥ [vɤo] Non-syllabic	ɇ̞̚ [vɤh] Advanced tongue root	
Ƀ [vɤr] Rhoticity	ɇ̞̚̚ [vh] Retracted tongue root	

VOWELS



Where symbols appear in pairs, the one to the right represents a rounded vowel.

SUPRASEGMENTALS

- ˈ Primary stress
- ˌ Secondary stress
- ː Long
- ˑ Half-long
- ɐ Extra-short
- | Minor (foot) group
- || Major (intonation) group
- Syllable break
- ⌋ ⌋ Linking (absence of a break)

TONES AND WORD ACCENTS

LEVEL	CONTOUR
ē [ɛ̄] Extra high	ě [ɛ̌] Rising
l̄ [ɛ̄˥] Extra high (tone letter)	
é [ɛ̌] High	ê [ɛ̋] Falling
l̇ [ɛ̌˥] High (tone letter)	
ē [ɛ̃] Mid	ē [ɛ̃ ɛ̃ ɛ̃] High rising
l̇ [ɛ̃˥] Mid (tone letter)	
è [ɛ̎] Low	ē [ɛ̎˥ ɛ̎˥] Low rising
l̇ [ɛ̎˥] Low (tone letter)	
è [ɛ̎˥] Extra low	ē [ɛ̎ ɛ̎ ɛ̎] Rising-falling
l̇ [ɛ̎˥˥] Extra low (tone letter)	
↓ [ɛ̎˥˥] Downstep	↗ [ɛ̎˥˥ ɛ̎˥˥] Global rise
↑ [ɛ̎˥˥] Upstep	↘ [ɛ̎˥˥ ɛ̎˥˥] Global fall